

Antoine Bonhomme

858-280-1736 | antoine.bonh@gmail.com | [linkedin.com/Antoine-Bonhomme](https://www.linkedin.com/antoin-bonhomme) | WEBSITE: [Antoine.engineer](https://www.antoine-engineer.com)

EDUCATION

University of California, Berkeley — B.S. Mechanical Engineering

Expected 2028

GPA: 3.95 | Relevant Coursework: Thermodynamics, Engineering Stats & Data Science, Internet-of-Things

EXPERIENCE

Berkeley Formula Racing - Autonomous Systems & Powertrain Member

Sep 2026 – Present

- **Autonomous (Controls):**
- Developing a Python-based path-following controller for racecar model, modeling kinematic bicycle dynamics and implementing feedback control with steering and acceleration constraints
- Developing centerline tracking, heading/lateral-error feedback, and trajectory control for safe, time-efficient lap completion
- **Powertrain (Mechanical):**
- Designed a differential mounting assembly and motor mount for a chain-driven rear drivetrain (EMRAX 228 motor, 3.53:1 reduction, Drexler FSAE diff), including bearings, fasteners, jacking bar, and chain tensioner
- Ran FEA in SolidWorks to a 1.5 minimum factor of safety under 4G shock loading

SeebYTE - Mechanical Engineering Intern

May 2026 – Aug 2026

- Retrofitted two legacy Bluefin Hovering AUVs (HAUVs) subsea robots built for autonomous dam-inspection missions, holding precise heading, depth, and position in high-current environments
- Developed a modernized control stack, integrating thrusters, a power distribution board, and a tethered Ethernet comms link
- Authored Python APIs to parse and fuse IMU, Doppler Velocity Log (DVL), and depth-sensor data into the vehicle's MAVLink-based flight controller, enabling closed-loop autonomous dynamic positioning
- Integrated a stereo camera and 3D sonar payload for autonomous underwater perception, and tuned PID controllers to achieve stable station-keeping, validated through pool testing
- Authored supporting hardware, software, and deployment documentation

MesaFSD (Full Self-Driving) Club - President & Founder

May 2025 – May 2026

- Founded and led a 20-engineer multidisciplinary team to design and build an autonomous racing go-kart end-to-end, placing 5th nationally in the Autonomous Karting Series
- Designed embedded control systems for autonomous actuation on high level and low level computers, integrating signal level-shifting and 5V logic relays to actuate 48V solenoids and brake master cylinder pumps
- Achieved an IP65 ingress-protection rating on a salvaged competition chassis by directing MIG-welding repairs and fabricating custom-sealed enclosures (TPU 95A gaskets, PETG/ABS housings) to protect embedded compute hardware
- Secured \$8,500 in funding for sensor and mechanical hardware through grant applications and corporate sponsor pitches

RESEARCH

DERConnect | UC San Diego

Research Assistant

Mar 2025 – June 2025

- Developed a real-time hardware-in-the-loop (HIL) simulation of a microgrid, integrating physical hardware with simulated systems to test autonomous control strategies for preventing feeder overloads.
- Implemented and validated real-time control and system-response logic, contributing to a research paper published with Clean Coalition.
- Built out the DERConnect compute infrastructure by installing 30+ compute nodes and designing/fabricating custom mechanical mounts and server racks using CNC machining and shop tools.

PROJECTS

Off-Road Go-Kart

Sept 2023 – Jun 2024

- Designed and fabricated a fully functional go-kart from scratch, integrating braking, throttle, steering, and drivetrain systems; modeled frame and component geometry in Fusion 360 to resolve alignment and weld-joint tolerances
- Cut, machined, and MIG-welded the chassis using a plasma cutting table, grinders, drill press, and saws - diagnosing and correcting fitment and weld-quality issues throughout the build

TECHNICAL SKILLS

Software & Design: Python, Fusion 360, SolidWorks, Git, OpenCV, KiCad, FEA, PID Tuning

Embedded & Sensors: Raspberry Pi, ESP32, Doppler Velocity Log (DVL), Sonar, GPS, Computer Vision, Encoders,

Fabrication: MIG Welding, Plasma Cutting, 3D Printing (PETG/ABS/TPU 95A), Soldering